ClaimClarity architecture

ClaimClarity answers questions about a customer's claims (837P) and remittances (835). Each customer's data sits in its own S3 prefix, under its own KMS key and IAM roles. Six AWS checks, the numbered diamonds, enforce that on every write and read. The main board shows the whole system, and drill-downs 00–04 zoom into each part.

The numbered diamonds on every board are the six permission checks. Each number means the same check on every board, and sits on the arrow it guards. The matrix below shows what each role can do in one customer's prefix.

Gates and access

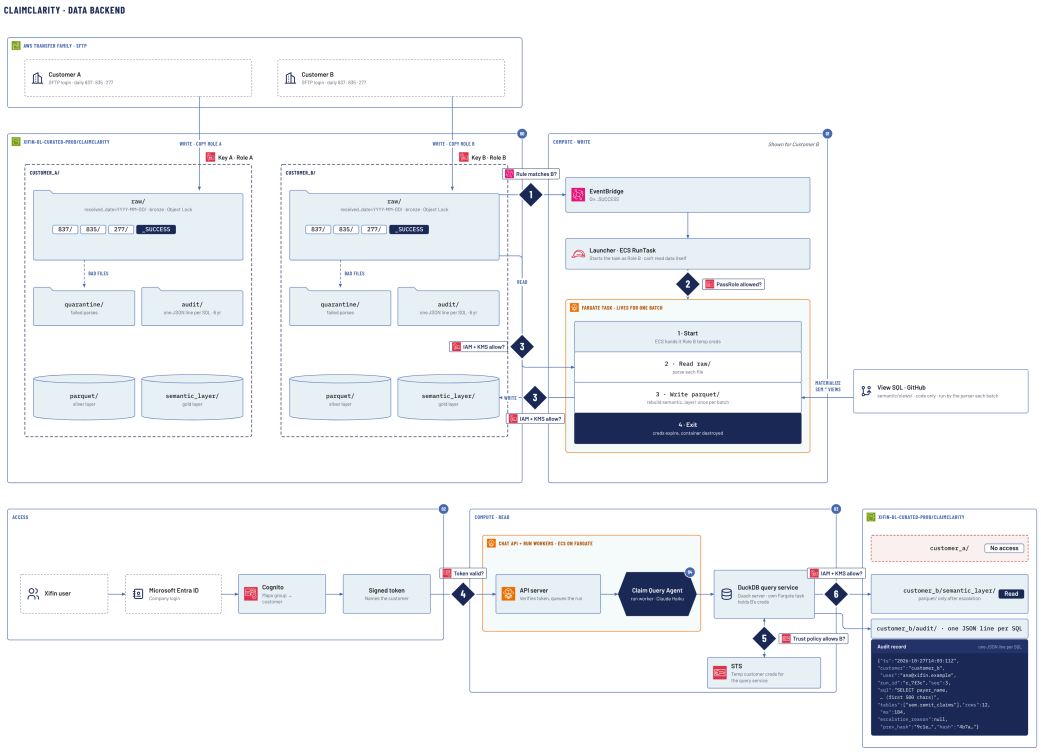
Gate	Checked by	What has to be true	If not	Boards
1	EventBridge rule	The _SUCCESS file is under this customer's `raw/`	No parser task starts	main, 01
2	IAM (`iam:PassRole`)	The launcher passes this customer's parser role, and only to an ECS task	The task can't start	main, 01
3	IAM + KMS	The parser role and the customer's key allow the read or write	The read or write fails	main, 00, 01
4	Chat API	The token is valid and names the customer	The request is refused	main, 02
5	STS (role trust policy)	The reader role trusts this token's customer	No credentials are issued	main, 03
6	IAM + KMS	The reader role and the customer's key allow the read	The read fails	main, 00, 03, 04

Access matrix, shown for customer B. Every customer has the same set of roles, each limited to its own prefix `customer_b/` and its own key.

Role (used by)	`raw/`	`quarantine/`	`parquet/`	`semantic_layer/`	`audit/`	Key	Gate
SFTP copy role B (Transfer Family)	Write once	–	–	–	–	Customer key B	–
Launcher (EventBridge → ECS RunTask)	–	–	–	–	–	–	1, 2 (may pass parser role B only)
Parser role B (Fargate parser task)	Read	Write	Write	Write a new version	–	Customer key B	3
Reader role B (DuckDB query service)	–	–	Read, only after escalation	Read the current version	–	Customer key B, decrypt only	5, 6
Audit role B (DuckDB query service)	–	–	–	–	Append, write once	Platform audit key	–
Chat API, run worker (Claim Query Agent)	–	–	–	–	–	–	4
Any customer A role	Denied	Denied	Denied	Denied	Denied	Denied	3, 6

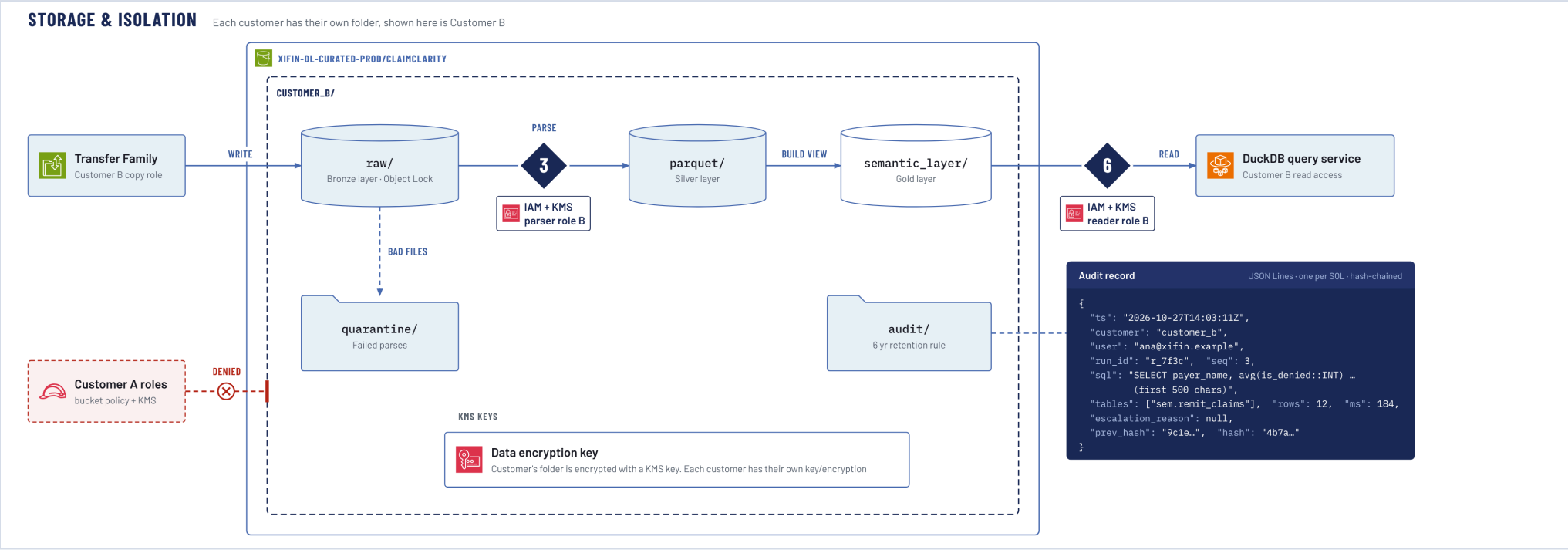
`raw/` and `audit/` are write-once (Object Lock): nobody can change or delete them, including the roles that write them. "Only after escalation" is enforced by the SQL guard; IAM allows the read. The chat API and the run worker hold no AWS data credentials at all.

Data backend



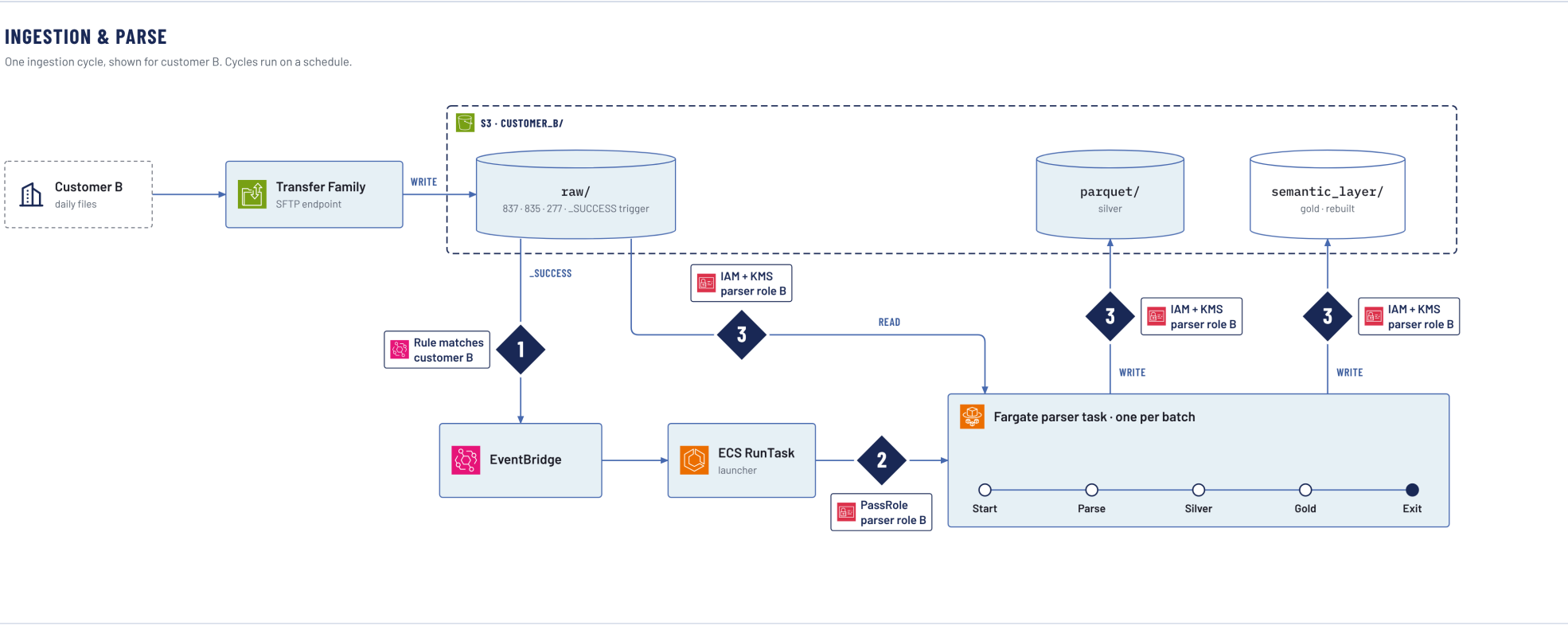
Part	What it does	Key facts
SFTP	Each customer sends daily 837 · 835 · 277 files	AWS Transfer Family, customer IP addresses only, straight into `raw/`
ClaimClarity data bucket	One prefix per customer: raw → silver → gold, plus quarantine and audit	One bucket for all customer data
Compute · write	EventBridge → ECS RunTask → one Fargate parser task per batch	Runs as the customer's parser role. Gates 1–3.
Access	Entra ID → Cognito → token naming the customer	The token is the only source of the customer. Gate 4.
Compute · read	Chat API → run worker (the Claim Query Agent) → DuckDB query service → STS → gold	15-minute credentials, this customer's gold only. Gates 5–6.

00 · Storage & isolation



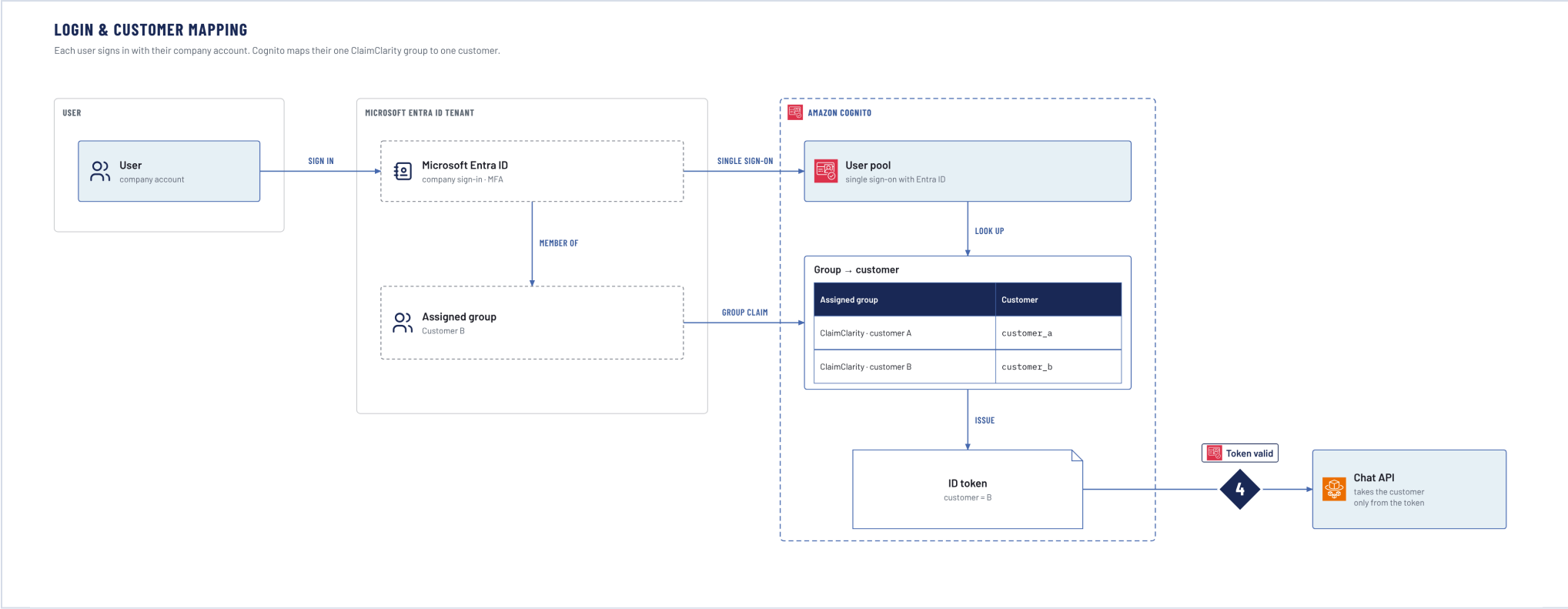
Part	Key facts
Bucket	One bucket for all customer data, one prefix per customer. The web app and the access logs have their own buckets.
Prefixes	`raw/` (write-once, Object Lock 7 years, `_SUCCESS` written last), `parquet/` (silver), `semantic_layer/` (gold), `quarantine/` (bad files, kept 1 year), `audit/`
IAM roles	Per customer: SFTP copy, parser, reader, audit. IAM prefix policies keep the roles apart.
KMS keys	The customer key, one per customer, covers raw, silver, gold and quarantine. The platform audit key covers `audit/`, with one key-policy statement per customer.
Bucket policy	Only customer B's roles reach `customer_b/`. Customer A's roles are denied by the bucket policy and the key policy.
Query audit	The DuckDB query service writes one JSON Lines record per SQL statement: write-once, kept 6 years, each record chained by hash to the one before
Gates	3 on the parser's reads and writes, 6 on the reader's reads

01 · Ingestion & parse



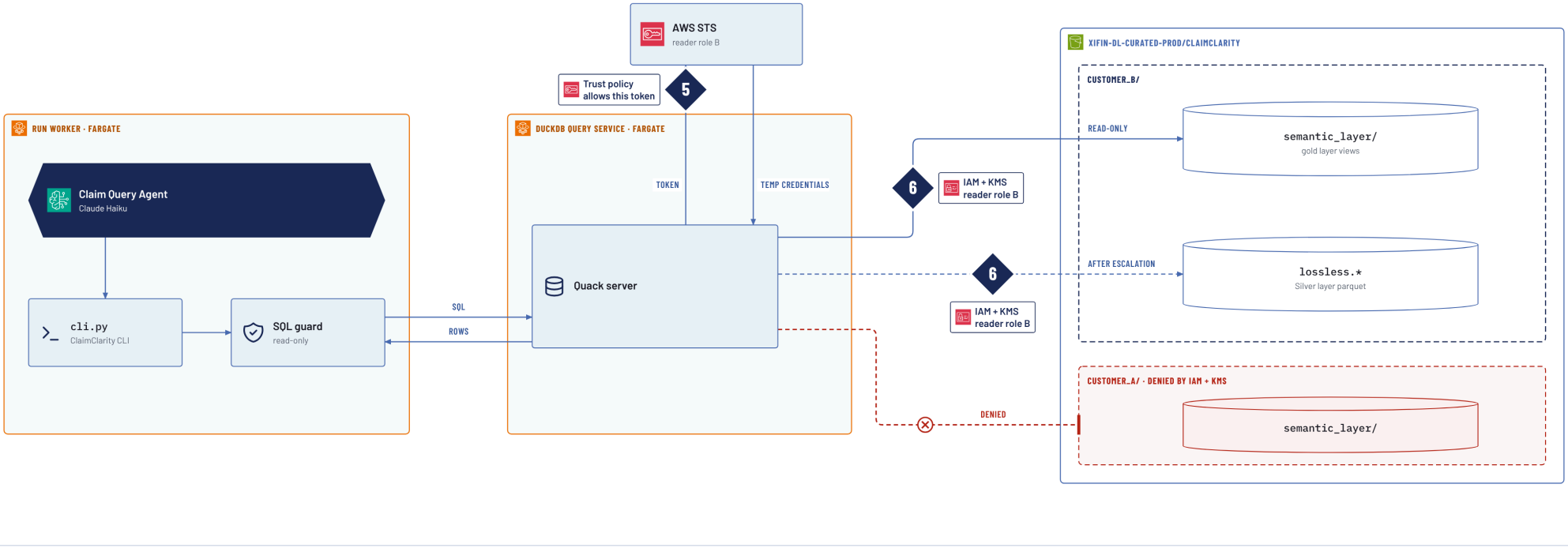
Step	What happens
Intake	The customer uploads over SFTP (port 22, customer IP addresses only). Transfer Family writes straight into `raw/` as the SFTP copy role, `_SUCCESS` last.
Gate 1	The customer's EventBridge rule matches `_SUCCESS` under `customer_b/raw/` only
Launch	ECS RunTask starts one parser task. EventBridge retries the launch 3 times.
Gate 2	`iam:PassRole` lets the launcher pass only customer B's parser role, and only to ECS tasks
Parser task	Fargate, 4–8 GB of memory, one per batch. It holds the batch lease, a lock object, while it runs. Bad files go to `quarantine/`.
Gate 3	IAM + KMS on every read of `raw/` and every write to silver and gold
Publish	Gold is written as a new version, and `CURRENT` moves by compare-and-set. Old versions stay readable for sessions already open.

02 · Login & customer mapping



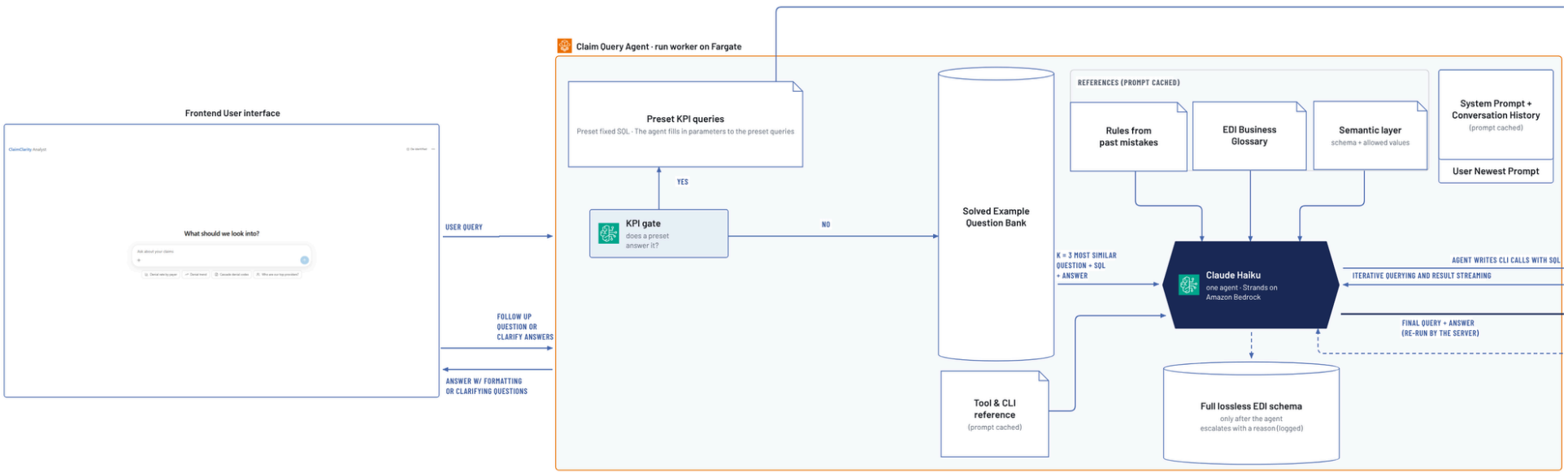
Part	Key facts
Microsoft Entra ID	Company sign-in. The ClaimClarity app is assigned to groups.
Assigned group	One group per customer
Amazon Cognito	Single sign-on with Entra ID. Maps the user's group to exactly one customer and issues a token naming that customer. A user in no ClaimClarity group, or in more than one, is refused.
Gate 4	The chat API accepts a request only with a valid token, and takes the customer only from it

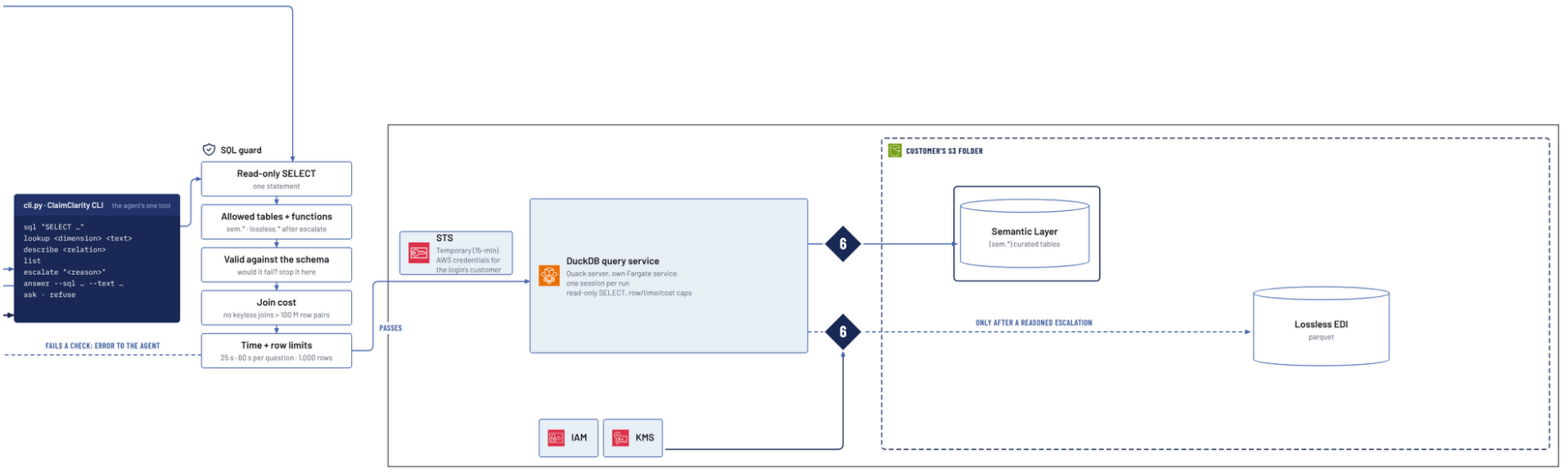
03 · Query path



Part	Key facts
Run worker	The Claim Query Agent sends SQL through `cli.py` and the SQL guard. It holds no AWS credentials.
SQL guard	Read-only SELECT. Blocks queries that would fail against the schema. 25 s per query, 60 s per question, 1,000 rows. Refuses keyless joins over 100 million row pairs.
Quack	SQL goes to the DuckDB query service over Quack with a per-run token, and rows come back the same way
Gate 5	The reader role's trust policy accepts only this token's customer. STS returns read-only credentials for 15 minutes, renewed with the same token when fewer than 3 minutes remain.
Gate 6	IAM + KMS allow reader role B only the current gold version in `customer_b/semantic_layer/`, and `customer_b/parquet/` after an escalation
`lossless.*`	PHI-stripped tables published next to gold, read only after the agent escalates with a reason
Customer A	Denied twice: the API takes the customer only from the token, and IAM + KMS deny reader role B on `customer_a/`

04 · Analyzer


04 · ANALYZER
The Claim Query Agent is the run worker. The DuckDB query service holds the credentials.
Frontend User interface
What should we look into?
USER QUERY
FOLLOW UP QUESTION OR CLARIFY ANSWERS
ANSWER W/ FORMATTING OR CLARIFYING QUESTIONS
Claim Query Agent · run worker on Fargate
Preset KPI queries
Preset fixed SQL · The agent fills in parameters to the preset queries
YES
KPI gate
does a preset answer it?
NO
Solved Example Question Bank
REFERENCES (PROMPT CACHED)
Rules from past mistakes
EDI Business Glossary
Semantic layer
schema + allowed values
System Prompt + Conversation History
(prompt cached)
User Newest Prompt
K = 3 MOST SIMILAR QUESTION + SQL + ANSWER
Claude Haiku
one agent · Strands on Amazon Bedrock
AGENT WRITES CLI CALLS WITH SQL
ITERATIVE QUERYING AND RESULT STREAMING
FINAL QUERY + ANSWER (RE-RUN BY THE SERVER)
Tool & CLI reference
(prompt cached)
Full lossless EDI schema
only after the agent escalates with a reason (logged)



Part	Key facts
KPI gate	One Claude Haiku call with structured output. If a preset fits, it fills the parameters (payer, payer class, month, CPT, CARC), which are checked against the data. Anything else goes to the agent.
Preset KPI queries	16 fixed SQL queries. They pass the SQL guard like any other SQL.
Claim Query Agent	The run worker itself (Fargate): Claude Haiku, Strands on Amazon Bedrock
Prompt	Cached: rules, EDI glossary, semantic schema with allowed values, tool and CLI reference, system prompt and history. Per question: the 3 most similar solved examples and the newest prompt.
First decision	Answer, ask a clarifying question, or escalate to the lossless tables with a reason, which is logged
`cli.py` · SQL guard	The agent's one tool: `sql`, `lookup`, `describe`, `list`, `escalate`, then `answer`, `ask` or `refuse`. Every query passes the guard's checks (limits in 03). A failed check goes back to the agent as an error.
Answer	Streams back with live steps. The server re-runs the final SQL, so the numbers come from the database.

Network, operations, log retention and session handling are owned by IT.

Open decisions

Question	Default shown
SFTP: AWS Transfer Family or an EC2 host?	Transfer Family, writing straight into `raw/`
Does the audit record keep the SQL text?	Yes, the first 500 characters
Queue and run workers	SQS + Fargate run workers
Per-customer limits	Concurrency and token budgets still need values